ADA University, SITE, Fall 2024, Discrete Structures

Midterm 1 (October 19, 2024). **SOLUTION SHEET**

27 points

Hint.

- Special sequences for the case of THREE input variables A, B, C (or x, y , z):

  A (or x) = 00001111, B (or y) = 00110011, C (or z) = 01010101

- The truth table for the minterms (**using the special sequences for x, y, z**) are as follows:

Table 1

Truth tables for minterms using the special sequences for x, y, z

xyz=00000001	xyz′=00000010	xy′z=00000100	xy′z′=00001000
x′yz=00010000	x′yz′=00100000	x′y′z=01000000	x′y′z′=10000000

Tasks of Midterm I

Task 1 (4 pt). Design a three-input **minimal** AND-OR circuit L with the following truth table:

T =[x, y, z; L]=[00001111, 00110011, 01010101; **10011011**]

Hint.
Step 1. Write truth table as a sum of minterms. (**1 pt**)
Step 2. Create a Karnaugh Map based on result of the Step 1. (**1pt**)
Step 3. Find an appropriate MSoP (**1 pt**)
Step 4. Design a three-input **minimal** AND-OR circuit L which corresponds to the Step 3 (**1 pt**)

Solution.

Step 1. (1 pt) Input sequences for the given circuit are special sequences. Represent the output sequence of bits generated by the given circuit, namely, **10011011** as a sum of minterms according to Table 1. We have **10011011**=x′y′z′+x′yz+xy′z′+xyz′+xyz

Step 2. (1 pt) Put checks in K-map according to the last expression.

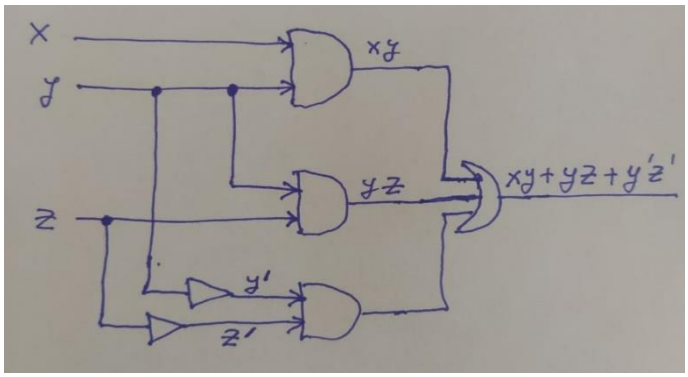
(A)	yz	yz′	y′z′	y′z
x	V	V	V	
x′	V		V	

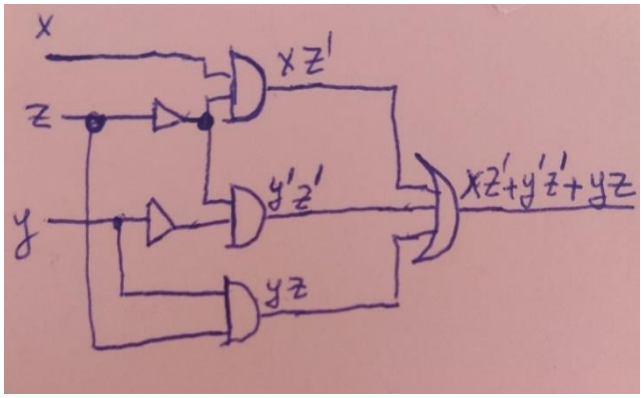
Step 3. (1 pt) MSoP= yz+y′z′+xy or MSoP=yz+ y′z′+ xz′

(It will be enough if the student suggests only one option).

Step 4. (1 pt) M**inimal** AND-OR circuit L which corresponds to is MSoP

(See next page)





MSoP= yz+y′z′+xy

MSoP= yz+ y′z′+ xz′

(It will be enough if the student suggests only one option).

Task 2 (5=3+1+1 pt).

Find the truth table T=T(E) for the Boolean expression

$$E=E(x, y, z)=(x'y+z)'(x'+y'z)+(x'z+y)(y'z)'+xz'+x'y'z$$

Hint.
Step 1. Find SoP form for E. (**3 pt**)
Step 2. Find CSoP form (**1 pt**)
Step 3. Use Table 1 to find T(E) (T(E) will be the sequence of eight digits 0 or 1). (**1 pt**)

Solution.

Step 1. (3 pt). Find SoP form for E.

E=E(x, y, z)=(x′y+z)′(x′+y′z)+(x′z+y)(y′z)′+xz′+x′y′z =

[(x′y)′z′(x′+y′z)]+[(x′z+y) (y+z′)]+xz′+x′y′z=

[(x+y′)z′(x′+y′z)]+[x′yz+x′zz′+yy+yz′]+xz′+x′y′z =

[xz′x′+ xz′y′z+y′z′x′+y′z′y′z]+[x′yz +y+yz′]+ xz′+x′y′z=

[x′y′z′]+[x′yz+y+yz′]+ xz′+x′y′z = x′y′+x′yz+xz′+y= **x′y′+xz′+y**

Step 2. (1 pt). Find CSoP form for E.

x′y′+x′yz+xz′+y=x′y′z+x′y′z′+ x′yz+ xyz′ + xy′z′+ xyz + xyz′ +x′yz + x′yz′=

= x′y′z+x′y′z′+ x′yz+ xyz′ + xy′z′+ xyz + x′yz′

Step 3. (1 pt). Find T(E)

T(E)= x′y′z+x′y′z′+ x′yz+ xyz′ + xy′z′+ xyz + x′yz′=0100 0000+1000 0000+0001 0000+
+0000 0010 + 0000 1000 + 00000001 + 00100000 =**1111 1011**

Answer: T(E)= **1111 1011**

Task 3 (4 pt=1+3)

Find a Boolean expression E=E(x, y, z) in **MSoP form** corresponding to the truth table T(E):

T(E)=10011011

Hint.

1. Find CSoP form for E by using Table 1. (**1 pt**)
2. Then use Karnaugh map method to reduce CSoP form to an MSoP form. It is enough to find at least one MSoP form if there are several such forms. (**3 pt**)

Solution.

Step 1. (1 pt). Find CSoP form for E by using Table 1.

CSoP(E)=x′y′z′+ x′yz + xy′z′ + xyz′+xyz

Step 2. (3 pt).

	yz	yz′	y′z′	y′z
x	V	V	V	
x′	V		V	

It is easy to see that

MSoP= yz+y′z′+xy OR MSoP= yz+ y′z′+ xz′

Task 4 (8=4+4 pt)

1.Find a MSoP for the Boolean expression E given by Table (A). (*Write as an expression*). **(3pt)**
2.Find a MSoP for the Boolean expression E given by Table (B). (*Write as an expression*). **(3pt)**

(A)	**zt**	**zt′**	**z′t′**	**z′t**
xy	V	V	V	V
xy′		V	V	
x′y′		V		
x′y	V	V	V	V

(B)	**zt**	**zt′**	**z′t′**	**z′t**
xy			V	V
xy′		V	V	
x′y′	V		V	
x′y	V		V	V

Solution.

MSoP=y+zt′+xt′

MSoP=yz′+ z′t′+x′zt+ xy′t′

Task 5 (3 pt)

Subtask 5.1 (1 pt). Find the negation of the proposition below

$\forall x \forall y \exists z P(x, y, z)$

Subtask 5.2 (2 pt). Domain of all predicates P(x) below is the set of *all integers*. What are the truth values of the corresponding propositions?

a) $\forall x(x>2)$ **(0.5 pt)** b) $\exists !x(x^2=1)$ **(0.5 pt)**

c) $\exists !x(x^2=-1)$ **(0.5 pt)** d) $\exists x(x=4x+1)$ **(0.5 pt)**

Reminder. Quantifier $\exists !$ means "There exists only one".

Solution.

Subtask 5.1 (1 pt).

$\neg(\forall x \forall y \exists z P(x, y, z)) = \exists x \neg(\forall y \exists z P(x, y, z) = \exists x \exists y \neg(\exists z P(x, y, z)) =$

$= \exists x \exists y \forall z(\neg P(x, y, z))$

Subtask 5.2 (2 pt).

a) False b) False c) False d)False

Task 6 (2 pt) Prove that every infinite set is equivalent to one of its proper subsets.

Proof. We know that any infinite set M contains a countable subset, say:

$A=\{a_1, a_2, ..., a_n, ...\} \subset M.$

Partition A into two countable subsets

$A_1=\{a_1, a_3, a_5, ...\}, \quad A_2=\{a_2, a_4, a_6, ...\}.$

Obviously, we can establish a **bijection** f between the countable sets A and A_1, $f:A \to A_1$ say, $a_n \leftrightarrow a_{2n-1}$, that is, $f(a_n)= a_{2n-1}$.

This bijection can be extended to a **one-to-one correspondence** between the sets $A \cup (M-A)$ and $A_1 \cup (M-A)$, by simply assigning x itself to each element $x \in M-A$.

But $A \cup (M-A)=M$, and $A_1 \cup (M-A)=M - A_2$. Hence M is equivalent to $M - A_2$ but $M - A_2$ is a proper subset of M. Statement is proved.

Task 7 (1 pt). Let S be a universal set and A and B are subsets of S. Prove De Morgan's law:

$(A \cup B)' = A' \cap B'$

Proof. Let $x \in (A \cup B)' \Leftrightarrow x \in S-(A \cup B) \Leftrightarrow x \notin (A \cup B) \Leftrightarrow (x \notin A \text{ and } x \notin B) \Leftrightarrow (x \in A' \text{ and } x \in B') \Leftrightarrow x \in (A' \cap B')$ ■

Good Luck!

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